FEDERATED GRID STABILIZATION : A RIDGE REGRESSION & VIRTUAL INERTIA APPROACH

Machine Learning

CSE 4622



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THE PROBLEM

- Global shift to renewable energy reduces spinning turbines in the grid
- Loss of physical rotational inertia in power systems
- Grid loses natural “shock absorber” effect
- Frequency becomes unstable during sudden load changes
- Fast frequency collapse risk increases significantly



THE RESEARCH GAP ADDRESSED

- Current smart inverters act only after frequency deviation occurs
- Reactive control leads to delayed response
- Grid instability often already begins before correction starts
- Cloud-based AI introduces communication latency
- Centralized decision-making is too slow for real-time grid control



MOTIVATION

- Need for stable renewable-powered grids
- Maintain frequency within safe limits
- Ensure real-time grid resilience
- Enable smooth renewable integration
- Prevent large-scale blackouts
- Build smarter, predictive grid control systems
- Transition toward fully sustainable power systems



Research Question

" How effectively can lightweight Time–Series Ridge Regression predict sub–cycle grid frequency transients at the edge? ”

“What extent does Federated Averaging (FedAvg) improve an isolated node's prediction accuracy for novel anomalies it has never locally experienced? "

RELATED WORK & THE IDENTIFIED GAP

What Was Done Before? (Prior Approaches & Limitations)

- Deep Reinforcement Learning (e.g., DDPG for Microgrids): Achieves high accuracy for grid management, but introduces massive computational overhead. Too heavy and slow for sub-cycle edge execution. [1]
- Traditional Virtual Inertia (e.g., PI/Droop Controllers): Successfully emulates physical inertia, but operates reactively. Suffers from phase-lag and lacks the predictive foresight to arrest frequency drops before they happen. [2]
- Federated Grid AI (e.g., FedForecast): Effectively preserves data privacy across distributed nodes, but currently focuses entirely on hourly or daily load forecasting, ignoring millisecond-level hardware stabilization. [3]

What Is New About Our Work?

- **Predictive Instead of Reactive** : We shift virtual inertia from a delayed reactive response to a preventative, predictive injection utilizing historical sliding windows.

- **Lightweight Mathematical AI** : We replace "black box" deep learning with closed-form Time-Series Ridge Regression, mathematically guaranteeing sub-millisecond execution on standard edge microcontrollers.

- **Millisecond Federated Control** : We adapt Federated Averaging (FedAvg) to share high-speed transient anomaly signatures across nodes, applying FL to ultra-fast hardware control rather than slow energy forecasting.

Dataset

Dataset Description

- Source & Validity: We utilize the PMU Dataset for Power Line Fault Detection (Kaggle), which is based on the IEEE 39-Bus New England Transmission System. This is a highly credible benchmark for grid stability research, widely accepted for simulating frequency and voltage transients. [5]

Dataset Description

```
Shape: (24654, 7)
Bus_ID             int64
Voltage          float64
Voltage_Angle    float64
Current          float64
Current_Angle    float64
Frequency        float64
Class_Label        int64
dtype: object
   Bus_ID   Voltage  Voltage_Angle   Current  Current_Angle  Frequency  \
0      26  0.905720      15.987394  0.916479      32.264385  49.855681
1      18  1.030443      15.805921  0.727791      89.558183  49.948551
2      29  0.936245      -4.616599  1.202986      46.546774  50.181598
3      14  0.995263       0.427471  1.126786     -20.773186  49.989729
4      37  1.066686     -13.607065  1.014143      41.988381  49.989626

   Class_Label
0            1
1            0
2            1
3            1
4            0
```

Dataset

Preprocessing & Engineering Plan:

1. Feature Engineering: We calculate the Rate of Change of Frequency (RoCoF) via numerical differentiation (dF/dt) across the time steps to capture the physical velocity of grid transients.

2. Cleaning: We apply Z-score normalization to all continuous features to ensure equal mathematical weighting and stability during the Ridge matrix inversion.

3. Transformation: We discard the binary Class_Label to formally shift the dataset architecture from discrete classification to continuous time-series regression.

4. Windowing: We apply a sliding-window technique k=10 to combine the telemetry from t-9 through t into a single, flattened feature vector used to predict the target frequency at t+1.

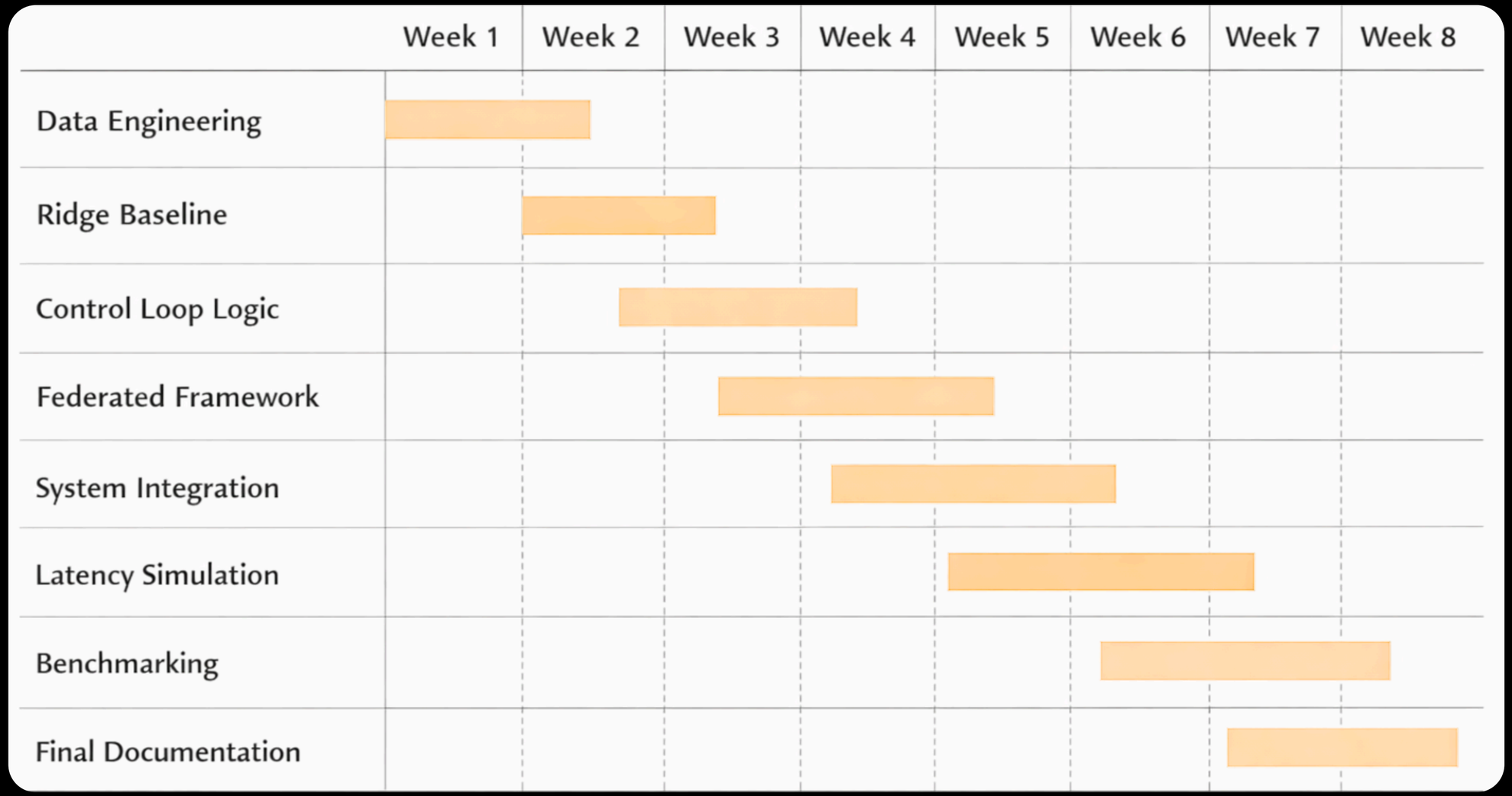
Expected Contributions

- A Novel Engineered Dataset: A continuous time-series regression dataset derived from standard PMU classification telemetry.
- A Lightweight Predictive Model: An edge-optimized, closed-form Ridge Regression pipeline executing sub-cycle frequency predictions.
- A Federated Simulation Tool: A multi-node Python framework simulating a decentralized Virtual Power Plant using a FedAvg server.
- A Comparative Benchmark Study: An empirical analysis proving predictive Ridge control outperforms traditional reactive baselines in both latency and frequency nadir (drop severity).

Tools

- Python: Core programming language.
- NumPy: For ultra-fast, low-level matrix inversion (crucial for executing the mathematical Ridge equation without heavy ML libraries).
- Pandas: For sliding-window dataset engineering and preprocessing.
- Git & GitHub: For version control and collaborative FedAvg server/client coding across the team.
- Matplotlib / Seaborn: For visualizing the predicted frequency trajectories vs. actual grid collapses.

TIMELINE


Week 1
Week 2
Week 3
Week 4
Week 5
Week 6
Week 7
Week 8
Data Engineering
Ridge Baseline
Control Loop Logic
Federated Framework
System Integration
Latency Simulation
Benchmarking
Final Documentation

Challenges and risk

- Overlapping Data: When data points are too similar, it confuses the system. The Fix: Clean and balance the data before teaching the model.

- Slow Connections: Bad internet in remote areas delays system updates. The Fix: Let each device update the main server independently whenever it has a connection, rather than waiting for everyone at once.

- Too Slow to React: Heavy, complex AI takes too long to make split-second decisions. The Fix : Use simpler, faster math methods so the system can react instantly.

References

1. Related Work (Literature)

[1] "A Deep Reinforcement Learning Framework for Adaptive Resiliency Enhancement in Smart Power Grids," IEEE Access.

[2] "Virtual Inertia Control-Based Model Predictive Control for Microgrid Frequency Stabilization Considering High Renewable Energy Integration," Sustainability (MDPI).

[3] "FedForecast: A federated learning framework for short-term probabilistic individual load forecasting in smart grid," International Journal of Electrical Power & Energy Systems.

[4]Ensemble learning based transmission line fault classification using phasor measurement unit (PMU) data with explainable AI (XAI)

2. Dataset

[5] New England IEEE 39-Bus System

3. Federated Framework & Methodology

[6] McMahan, B., et al. (2017). "Communication-Efficient Learning of Deep Networks from Decentralized Data," Artificial Intelligence and Statistics (AISTATS).

THANK YOU

Any Questions?